BLUE FAIRY DEVELOPMENT

BlueFairy Brew Tracker

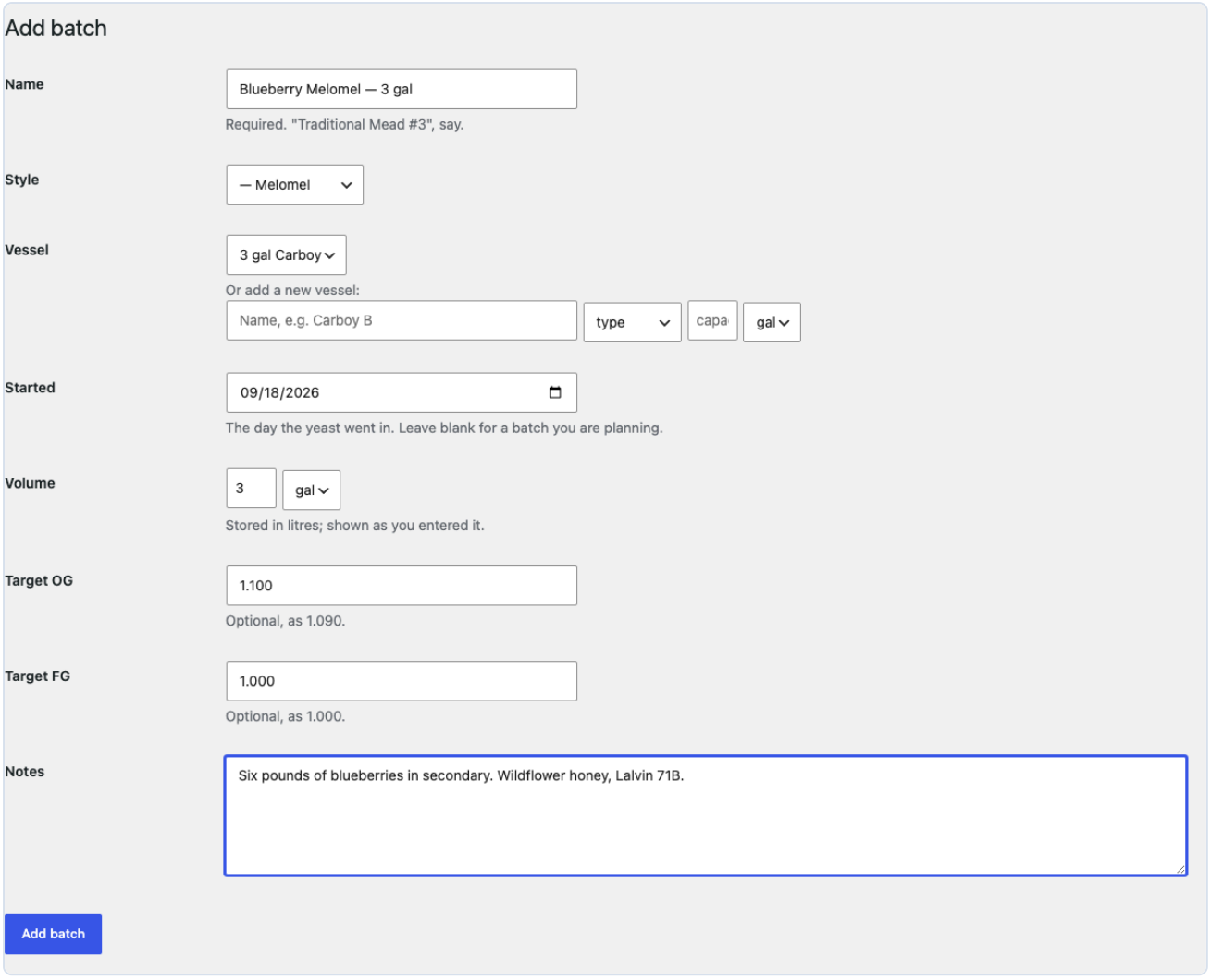
A batch log for mead, wine, cider and beer — gravity, ABV, timeline, tasks

Version 1.0.4 · GPL-2.0-or-later · WordPress 6.5+ · PHP 8.1+
bluefairydevelopment.com

BlueFairy Brew Tracker — features

Every statement here is a claim about the shipped code. If one stops being true, the code is wrong or this file is.

The batch record



Adding a batch: name, style, vessel, start date, volume, target OG and FG

One record, `bfbt_batch`, owned by this plugin. The Pro add-on extends the same record through `BFBT_Batch::register_field()` rather than keeping its own, which is why installing Pro shows you every batch you have already logged.

Field	Where it lives
Name	the post title
Style	a term in the `bfbt_style` taxonomy
Vessel	a term in the `bfbt_vessel` taxonomy, with its type and capacity as term meta
Started	`_bfbt_started` — the day the yeast went in; blank for a batch you are planning
Volume	`_bfbt_volume_l` in litres, with `_bfbt_volume_entered` keeping "1 gal" as you typed it
Target OG, target FG	`_bfbt_target_og`, `_bfbt_target_fg`
Racked to secondary, packaged, completed	`_bfbt_secondary_date`, `_bfbt_packaged_date`, `_bfbt_completed_date`
Dumped	`_bfbt_dumped` — `1`, or absent
Original gravity	`_bfbt_og_log_id` — which reading counts as OG
Notes	the post content, through `wp_kses_post`

The style tree is seeded on activation: Mead (Traditional, Melomel, Cyser, Metheglin, Braggot, Pyment), Wine (Red, White, Rosé, Fruit, Sparkling), Cider (Dry, Sweet, Perry), Beer (Ale, Lager, Stout, IPA, Wheat, Sour) — twenty-four terms. Seeding is by slug and never overwrites a term, so activating twice seeds once and a style you renamed keeps its name.

A vessel is a term with a type (carboy, bucket, demijohn, conical, keg, barrel, other) and a capacity. You pick an existing vessel from a list or type a new one on the batch form, so "Carboy A" stays one vessel however many batches go through it.

Check your batches from the command line:

```
wp post list --post_type=bfbt_batch --fields=ID,post_title,post_status
wp post list --post_type=bfbt_log --post_parent=<batch id> --fields=ID,post_title,post_date
wp post list --post_type=bfbt_task --post_parent=<batch id> --fields=ID,post_title
wp term list bfbt_style --fields=term_id,name,slug,parent
wp term list bfbt_vessel --fields=term_id,name
```

Validation refuses, never guesses

Every value is checked before anything is written and refused with the value quoted back. A refused save is never reported as a success.

- **Dates** are calendar-checked in year-month-day. `2026-02-30` has the right shape and is not a day, so it is refused. `03/14/2026` is refused rather than read as the 3rd of February.
- **Gravity** must be typed as `1.090`, in `0.980` to `1.200`. `1090`, `90` and a value that looks like Brix are refused with the hint "enter as 1.090" rather than reinterpreted into a reading you did not take.

- **Quantities** are non-negative decimals with a dot. **Units** come from a fixed list: g, kg, oz, lb, ml, l, gal, tsp, tbsp, pkt, tablet, each.
- **Temperature** is typed in °C or °F — a select beside the field, opening on the unit your setting implies — and checked in that unit: −10 to 60 °C, or 14 to 140 °F. The refusal names the unit you typed in (""200" is outside the temperature range this plugin accepts (14 to 140 °F)"). A °C entry above 60 that reads like an ordinary °F number — 74, say — is refused with the hint "If you meant Fahrenheit, pick °F next to the field or set your units to US in Settings." It is never converted for you.
- **pH** is 0 to 14.
- **Volume** is a number and a volume unit (ml, l, gal) and must be more than zero.

Your input survives a refused save

A refused save does not send you back to an empty form. Every admin form — add and edit batch, add entry, add task, milestones — comes back filled with what you typed, the refused value included so you can see what was wrong, and the entry form reopens on the type you submitted. The copy is kept for thirty seconds in a per-user transient beside the notice and dropped the first time the form reads it: a reload after that is a fresh form.

Units

Volumes are stored in litres, mass in grams, temperature in °C. One setting, **Display units** — "US (gallons, ounces, °F)" or "Metric (litres, grams, °C)" — decides two things: how a *derived* volume is shown (the batch size, say, appears as "1 gal (3.79 l)" in metric and "1 gal (1.00 gal)" in US), and which unit every form offers first (gal, oz and °F under US; l, g and °C under metric). A value you logged is never converted: a 1.2 g Fermaid-O dose is shown as 1.2 g whatever the setting says, and a reading typed as 74 °F stays "74 °F" — under metric it reads "74 °F (23.3 °C)", the same reading in the unit you now work in. The unit you pick on the entry is the one that counts.

The timeline

Every entry is a `bfbt_log` post with the batch as its parent, dated when it happened rather than when it was typed (the "When" field, `Y-m-d H:i`, calendar-checked). Seven entry types: addition, gravity, temperature, pH, note, racking, stage.

Type	What it records	Generated title
Addition	ingredient as typed, quantity, unit	`Fermaid-O 1.2 g`
Gravity	SG, and a temperature if you took one, in the unit you typed	`SG 1.042 @ 20 °C`, `SG 1.050 @ 68 °F`
Temperature	°C or °F as typed; stored in °C as well	`74 °F`
pH	pH	`pH 3.40`
Racking	optionally the volume that came across	`Racked — 3.5 l across`
Note	free text	the first sixty characters
Stage	a label	the label

Entries are immutable in this plugin. There is no edit form and the add handler refuses any posted entry id. A typo is trashed and re-entered, so a number on the batch screen can always be traced to the entry that produced it. Pro adds an **Edit** control on every row (fields, note, time and photos); an entry edited that way carries an *edited* mark with the time of the edit, so the trace still holds. The seam is `BFBT_Log::update()`, which runs the same validation as a new entry, never changes an entry's type or the task it completes, and stores `_bfbt_edited` / `_bfbt_edited_by`.

The timeline shows fifty entries a page, newest first. Trash is offered only to users who may delete others' posts.

An entry dated ahead of the site clock stays published. WordPress would otherwise schedule it as a future post and drop it from every query.

Photos on an entry

The entry form takes up to five images with the entry — the krausen, the hydrometer, the colour after a racking. Each becomes an ordinary media-library attachment whose parent is the entry, shown as a thumbnail in the timeline's Photos column (the thumbnail opens the full image) and named by URL in the CSV. Only image types the site accepts are stored; anything else is refused by name while the entry and the other photos are kept, because a reading typed in is worth more than a picture of it: "Entry added, but 1 photo was not saved. “notes.txt”: Sorry, you are not allowed to upload this file type." More than five refuses the whole entry before anything is written.

Photos follow the entry. Trashing the entry (or its batch) leaves them in place for a restore; deleting it permanently deletes them, row and file, where WordPress on its own would only orphan them. Uninstall keeps them, as it keeps the entry. A browser never re-fills a file input, so a refused save gives back every typed field but the photos; the form says so.

Original gravity, ABV and attenuation

Traditional Mead #3 Secondary Edit All batches

Traditional · Carboy A · 1 gal (1.00 gal) · started 2026-08-04 · 20 days in Secondary

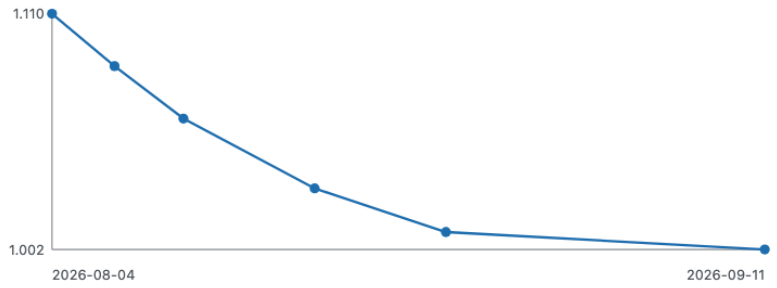
ABV 15.6 % · apparent attenuation 98 %

OG 1.110, reading of 2026-08-04 · FG 1.002, reading of 2026-09-11

76.08 × (OG − FG) / (1.775 − OG) × (FG / 0.794), the high-gravity formula for OG above 1.070

Orange blossom honey, 3 lb to the gallon. Staggered Fermaid-O, degassed twice daily for the first week.

Gravity



Due

2026-09-16	Gravity check — confirm stable	Done
2026-09-19	Potassium metabisulphite 1 tablet	Done

Done in the last 7 days

2026-09-18	~~Potassium sorbate 0.5 tsp~~	Undo

Add entry

Type: Addition · When: 2026-09-18 16:57

Ingredient: Fermaid-O

Quantity: g

Note

Add entry

Add task

Type: Addition

Ingredient: Fermaid-O

Quantity: g

Leave the quantity blank and put the instruction in the title — "water to 1 gal".

Title (optional)

Due: mm/dd/yyyy — Blank for a task with no date.

Add task

Timeline

When	Type	Entry	Note	
2026-09-18 16:56	Addition	Potassium sorbate 0.5 tsp		Trash
2026-09-11 10:00	Gravity	SG 1.002 @ 19 °C		Use as OG Trash
2026-08-29 11:00	Racking	Racked — 3.5 l across	Racked off the lees into a clean carboy.	Trash
2026-08-25 10:00	Gravity	SG 1.010		Use as OG Trash
2026-08-18 10:05	pH	pH 3.40		Trash
2026-08-18 10:00	Gravity	SG 1.030 @ 20 °C		Use as OG Trash
2026-08-14 08:00	Temperature	20 °C		Trash
2026-08-11 10:10	Gravity	SG 1.062		Use as OG Trash
2026-08-11 10:00	Addition	Fermaid-O 1.2 g		Trash
2026-08-07 18:00	Gravity	SG 1.086 @ 22 °C		Use as OG Trash
2026-08-06 10:00	Addition	Fermaid-O 1.2 g		Trash
2026-08-05 10:00	Addition	Fermaid-O 1.2 g		Trash
2026-08-04 10:05	Addition	Fermaid-O 1.2 g		Trash
2026-08-04 10:00	Gravity	SG 1.110 @ 21 °C		OG Trash
2026-08-04 09:45	Addition	Lalvin 71B 1 pkt		Trash
2026-08-04 09:30	Addition	Orange blossom honey 3 lb		Trash

Milestones

Racked to secondary: 08/29/2026

Packaged: mm/dd/yyyy

Completed: mm/dd/yyyy

☐ Dumped

Save milestones

The first gravity reading you log is pinned as the original gravity. Any other gravity reading on the timeline has a **Use as OG** button; trashing the pinned reading clears the pin.

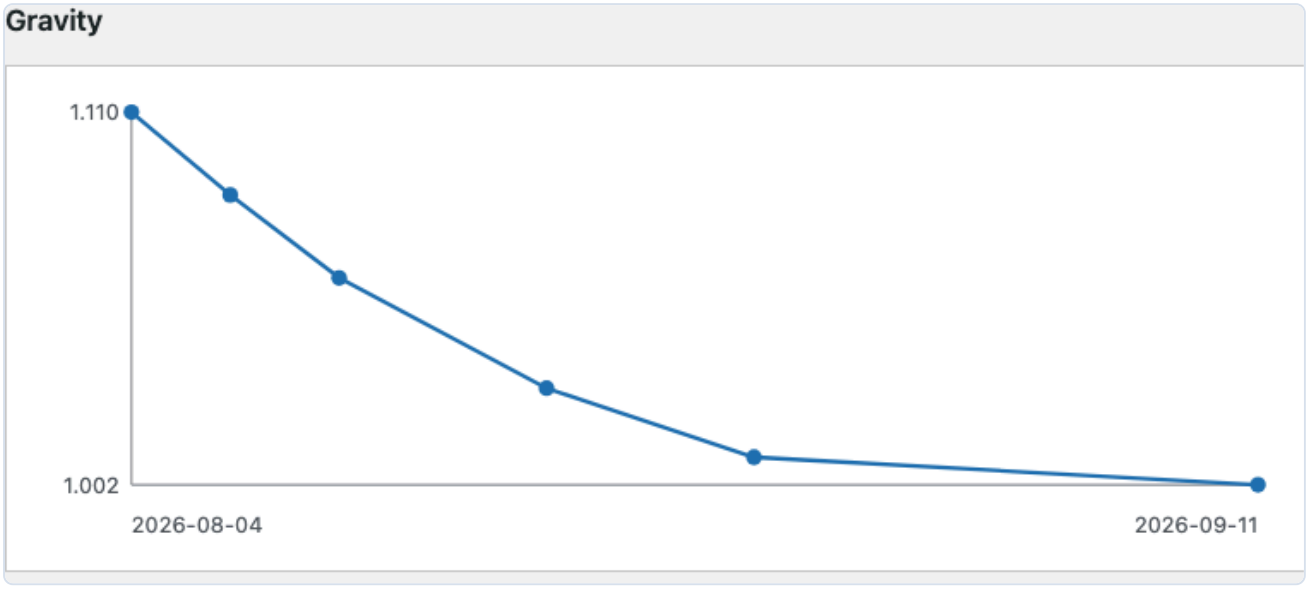
ABV and apparent attenuation are worked out at read time from the pinned OG and the latest gravity reading. Neither number is stored, and both are shown with their provenance — "OG 1.110, reading of 2026-03-01 · FG 1.000, reading of 2026-03-21" — so trashing or re-dating an entry changes a visible number, never a hidden one.

Two formulas, and the screen names the one it used:

OG	Formula
up to 1.070	`(OG − FG) × 131.25`
above 1.070	`76.08 × (OG − FG) / (1.775 − OG) × (FG / 0.794)`

The standard formula under-reports a mead by around 1.4 % ABV, which is most of the difference between a session mead and a sack mead.

The gravity chart



Every gravity reading over time, drawn by the plugin as inline SVG

An inline SVG of every gravity reading over time, up to five hundred points, drawn by the plugin with no JavaScript library and no external asset. Hovering a point shows the reading and its date. Every number is cast to a float and every label is escaped before it reaches the markup. A batch with no readings says "No readings yet."

Stage

Derived from the milestone dates, never stored, so a batch cannot disagree with itself.

Stage	When
Dumped	the batch is marked dumped
Complete	a completed date is recorded
Packaged	a packaged date is recorded
Secondary	a racked-to-secondary date is recorded
Primary	a start date is recorded
Planned	nothing yet

Most final wins: a batch that was completed and then poured away is dumped, and the completed date left behind does not hide that. The batch screen says how many days it has been in its stage.

Pro replaces the answer through the `bfbt_batch_stage` filter and inserts its own states into the order through `bfbt_stage_order`. It supplies only the states between the ends; planned, complete and dumped are decided here, once.

Tasks and what is due

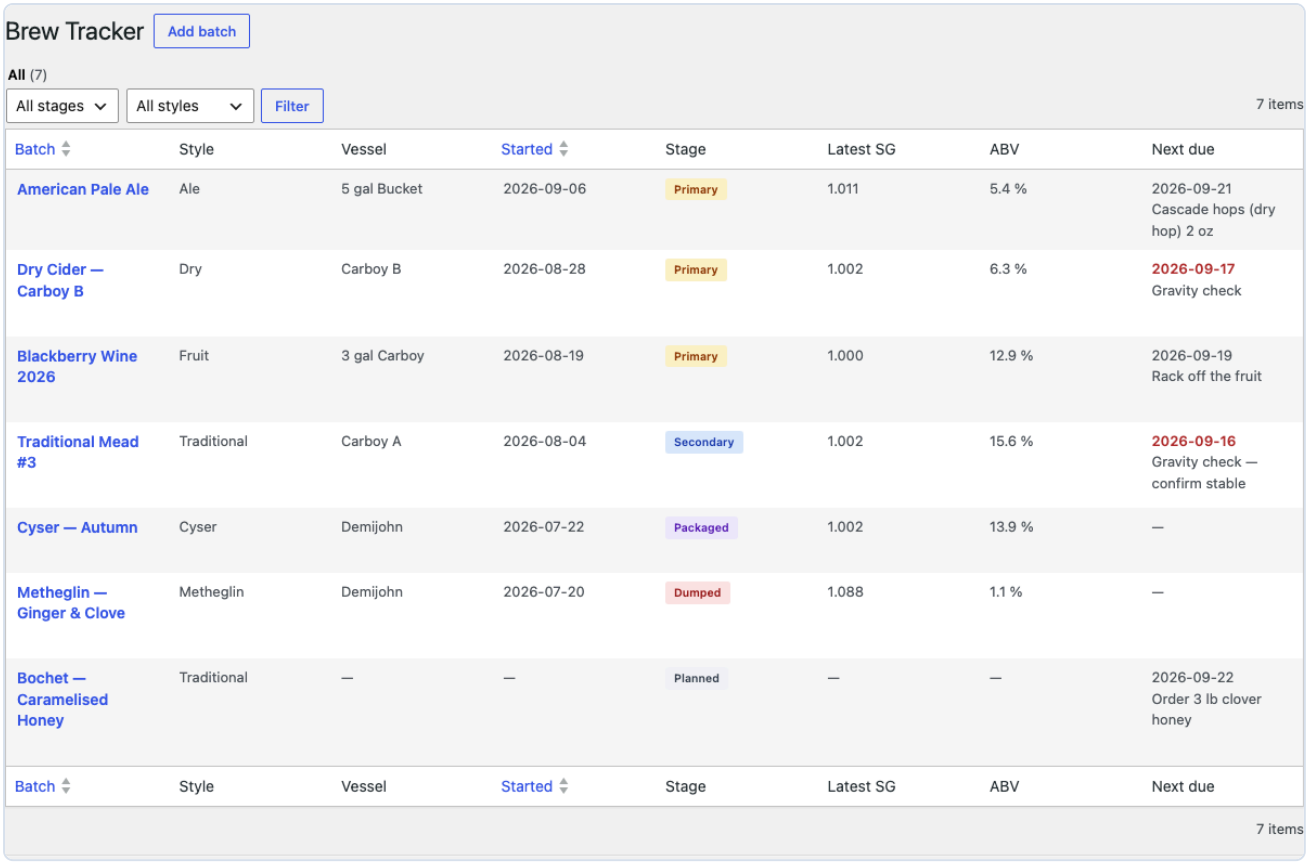
A task is a `bfbt_task` post: what to do, when, and when it was done. Five types: addition, gravity check, rack, package, other. An addition task carries an ingredient, quantity and unit, so "Fermaid-O 1.2 g" is one line and one button; leave the quantity blank and the instruction lives in the title ("water to 1 gal").

The **Due** panel on the batch screen lists open tasks soonest first — overdue in red, today in amber, undated in grey. **Done** logs the addition (or a note "Done: rack") to the timeline and stamps the task; **Undo** reopens it and leaves the entry in place. Completing is idempotent: a double-click or replayed form cannot log the same task twice.

The **Next due** column on the batch list shows each batch's earliest open task, red when overdue, worked out in one grouped query for the whole page. A task with no date never becomes "next due".

Tasks are typed by hand. Nothing here generates a schedule; that is the first thing Pro adds.

The list

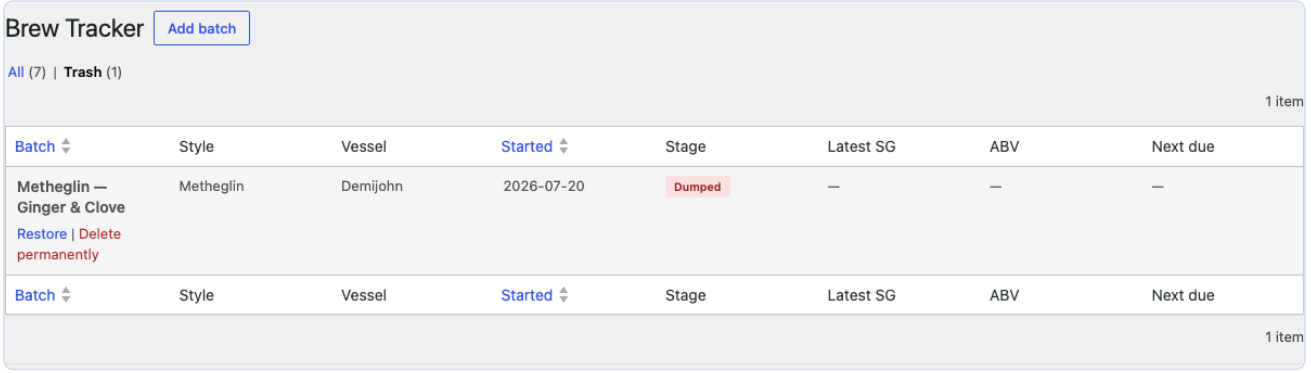


Every batch with its stage, latest gravity, ABV and what is due next

Twenty-five batches to a page, sorted and paged **in the database**. Columns: Batch, Style, Vessel, Started, Stage, Latest SG, ABV, Next due. Sort by name or by start date in either direction — a planned batch with no start date stays on the list. Filter by stage and by style.

All | **Trash** views. Trashing a batch takes its entries and tasks with it; restoring brings back exactly the ones that went with it, not an entry you had trashed on purpose beforehand; deleting permanently from the Trash view removes the batch and its children for good.

Who may do what

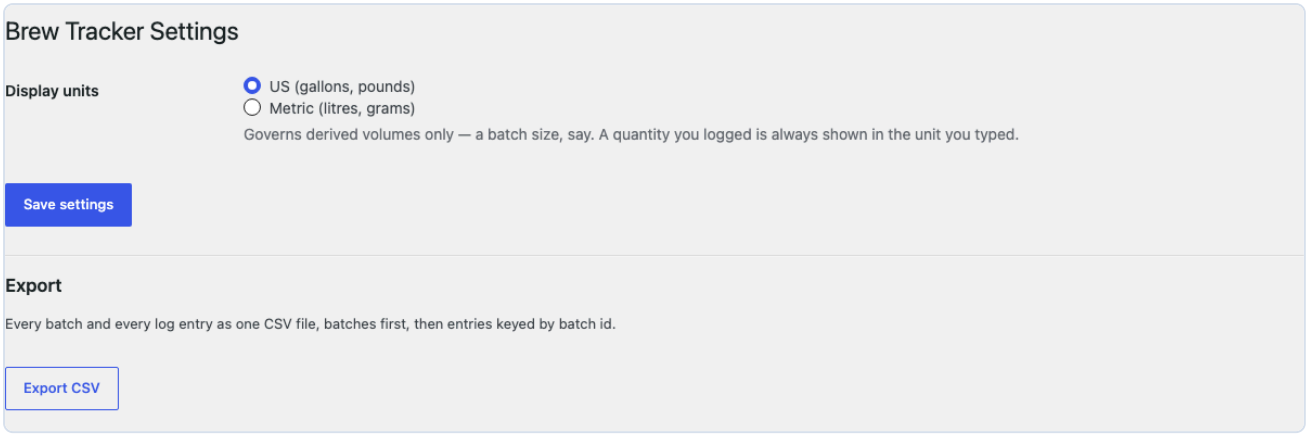


The Trash view: Restore brings back the batch with its entries and tasks; Delete permanently removes them for good

Action	Capability
See the screen, add and edit batches, log entries, add and complete tasks, pin OG	`edit_posts`
Trash, restore and delete a batch; trash an entry	`delete_others_posts`
Settings and CSV export	`manage_options`

A Contributor can log; only someone who may delete others' posts can destroy. Every write is a nonced form post to an `admin_post_*` handler that checks its capability first and confirms the post type of every id it is handed.

CSV export



Settings: display units, and the CSV export of every batch and entry

Settings → **Export CSV** streams every batch and then every entry as one file, five hundred rows at a time, never built in memory. Every field is quoted, and a value beginning with `=`, `+`, `-`, `@`, a tab or a carriage return is prefixed with a single quote so a batch named `=HYPERLINK(...)` opens in a spreadsheet as text, not as a formula. An entry's photos are named by URL in the last column of the entries section, `photos`, space-separated.

Uninstall

Uninstalling deletes `bfbt_features`, `bfbt_units`, `bfbt_seeded`, the plugin's per-user transients (the one-shot notice and a refused form's remembered input), and `bfbt_style_children` — the cached parent/child map WordPress itself keeps for the style taxonomy — never a batch, an entry, a task, a photo, a style or a vessel. There is no `LIKE 'bfbt_%'` sweep: that pattern would also match the Pro add-on's `bfbtp_` settings.

Extension points

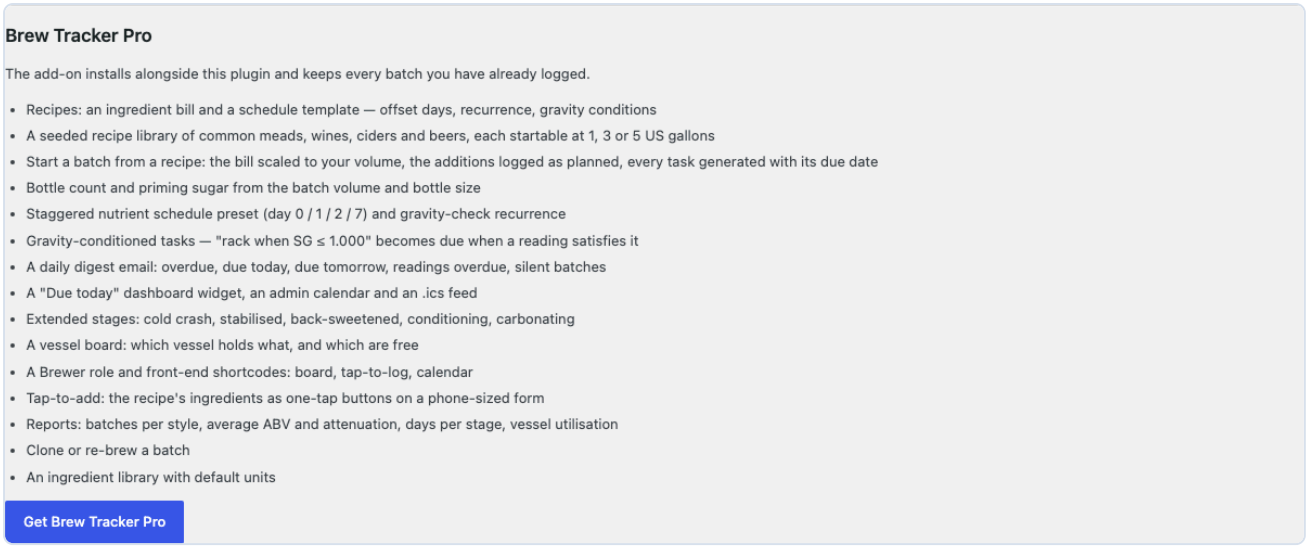
Hook	Purpose
`bfbt_loaded`	fires before this plugin registers its own subsystems; where Pro attaches
`BFBT_Batch::register_field()`	add a field to the batch record
`bfbt_batch_stage`, `bfbt_stage_order`, `bfbt_terminal_stages`	replace the derived stage, insert states, name the stages after which nothing is scheduled
`bfbt_stage_labels`, `bfbt_stage_colours`	name and colour new states
`bfbt_log_types`, `bfbt_task_types`	add entry and task types
`bfbt_batch_created`, `bfbt_batch_updated`, `bfbt_log_created`, `bfbt_task_completed`	react to a change
`bfbt_batch_form_fields`, `bfbt_batch_form_submit`	add fields to the add/edit form and read them back
`bfbt_batch_actions`	add buttons to the batch screen
`bfbt_log_row_controls`	add controls at the start of a timeline row's last cell (Pro's Edit link)
`bfbt_log_updated`	react to an entry rewritten through `BFBT_Log::update()`
`bfbt_after_batch_list`, `bfbt_settings_after`	add content below the list and the settings
`bfbt_admin_menu`	add pages under the Brew Tracker menu
`bfbt_ingredient_label`	resolve an ingredient typed as text
`bfbt_units`	override the display-unit setting
`bfbt_feature_labels`, `bfbt_feature_descriptions`	declare feature switches

Signatures are in `README.md`.

What this plugin does not do

- No schedule. It works out ABV and attenuation from what you logged, and nothing else; every task is one you typed.
- No email, no cron, no scheduled work, no dashboard widget.
- No REST routes, no front end, no database tables of its own.
- Three options of its own, all removed on uninstall, along with the style hierarchy cache WordPress keeps for it.

Free and Pro



What the paid add-on adds, shown beneath your batch list

This plugin is the base of a pair. Pro is an add-on: it extends this plugin rather than replacing it, so there is one batch record, one place your data lives, and nothing to migrate when you upgrade.

Pro column describes Brew Tracker Pro 1.0.

	Pro — $25	Free
Batch record: name, style, vessel, start date, volume, target OG/FG, notes	Yes	Yes
Style taxonomy seeded with Mead, Wine, Cider, Beer and their substyles	Yes	Yes
Vessel taxonomy with type and capacity	Yes	Yes
Timeline log per batch: additions, gravity, temperature, pH, notes, racking	Yes	Yes
Up to five photos on an entry, thumbnails on the timeline	Yes	Yes
Edit a timeline entry in place: fields, note, time, photos	Yes	—
ABV and attenuation from the pinned OG and the latest reading, both formulas stated	Yes	Yes
Gravity chart per batch, inline SVG	Yes	Yes
Six-stage cycle: planned, primary, secondary, packaged, complete, dumped	Yes, extended	Yes
Manual tasks with due dates, a Next due column and a Due panel	Yes	Yes
Paged, sortable batch list with a Trash view and Restore	Yes	Yes
CSV export of batches and entries	Yes	Yes
Batches kept on uninstall	Yes	Yes
Recipes — ingredient bill and schedule template	Yes	—
Seeded recipe library of common meads, wines, ciders and beers at 1, 3 or 5 US gallons	Yes	—
Start a batch from a recipe — bill scaled, additions planned, tasks generated	Yes	—
Bottle count and priming sugar from batch volume and bottle size	Yes	—
Staggered nutrient preset (day 0 / 1 / 2 / 7) and gravity-check recurrence	Yes	—
Gravity-conditioned tasks — "rack when SG ≤ 1.000"	Yes	—
Daily digest email — overdue, due today, due tomorrow, silent batches	Yes	—
Dashboard widget, admin calendar and .ics feed	Yes	—
Extended stages — cold crash, stabilised, back-sweetened, conditioning, carbonating	Yes	—
Vessel board — which vessel holds what, which are free	Yes	—
Brewer role and front-end shortcodes — board, tap-to-log, calendar	Yes	—
Reports — batches per style, average ABV, days per stage, vessel use	Yes	—
Clone or re-brew a batch	Yes	—
Ingredient library with default units	Yes	—

Why the missing rows cost you

You are keeping the schedule in your head. This plugin records the tasks you type. A one-gallon mead wants Fermaid-O on day 0, 1, 2 and 7, a gravity reading on day 3, 7 and 14, and a rack when the gravity settles — and you are the one counting the days for every carboy on the shelf. Pro keeps that as a recipe and "Start batch from recipe" creates every task with its date.

Nothing tells you until you look. The Due panel is on the batch screen, so you find out what is due when you open it. Pro sends a daily digest — overdue, due today, due tomorrow, batches that have gone quiet — and puts the same list on the dashboard, a calendar and a feed your phone subscribes to.

A rack is a judgement, not a date. "Rack to secondary when SG is at or below 1.000 and stable" cannot be typed as a task with a date. Pro's gravity-conditioned tasks become due when a reading satisfies them, and only when the last few readings agree.

You are scaling the bill by hand. Three pounds of honey for a gallon; how much for three? How many 16 oz bottles from five gallons after losses? Pro scales the recipe to the batch, keeps one yeast packet as one packet, rounds tablets up, and puts the bottle count on the packaging task.

Nobody else can log a reading. This plugin is a wp-admin screen. Pro adds a Brewer role and a phone-sized front end where the recipe's ingredients are one-tap buttons.

Get Pro

https://wordpress.bluefairydevelopment.com/product/bluefairy-brew-tracker-pro/

One payment, no subscription, lifetime licence for one site. Pro requires this plugin to be installed and active — it is an add-on, not a replacement, and every batch you have already logged carries straight over.